

| Question | Answer                                                                                                                                                                 | Marks     |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 4(a)     | (motion in which) acceleration is (directly) proportional to displacement                                                                                              | <b>B1</b> |
|          | (motion in which)<br>acceleration is (always) in the opposite <u>direction</u> to displacement<br><b>or</b><br>acceleration is (always) directed towards a fixed point | <b>B1</b> |
| 4(b)(i)  | $t_1$ <b>and</b> $t_5$<br><br><b>or</b><br><br>$t_3$ <b>and</b> $t_7$                                                                                                  | <b>B1</b> |
| 4(b)(ii) | $f = 1 / T$                                                                                                                                                            | <b>C1</b> |
|          | period = $2.2 / 1.25$<br><br>$f = 1.25 / 2.2$<br><br>= 0.57 Hz                                                                                                         | <b>A1</b> |
| 4(c)     | $v_0 = \omega x_0$ <b>and</b> $\omega = 2\pi f$                                                                                                                        | <b>C1</b> |
|          | $0.080 = 2\pi \times 0.57 \times x_0$                                                                                                                                  | <b>C1</b> |
|          | $x_0 = 0.022 \text{ m}$                                                                                                                                                | <b>A1</b> |

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